COMPUTER SCIENCE · NETWORKING NOTES · REVISED & EXPANDED EDITION

Data Communication & Computer Networks

A complete, exam-ready unit reference

Data Communication Fundamentals · OSI & TCP/IP Models · Transmission Media · Network Topologies · Multiplexing (FDM / TDM / WDM) · Circuit & Packet Switching · LANs & Interconnection Devices · PDU Encapsulation

72 numbered sections · hierarchical sub-numbering · worked examples and extra notes added throughout for deeper understanding · corrections highlighted

Table of Contents

1. Data Communication Concepts
2. Characteristics of Data Communication
3. Data Flow (Simplex / Half-Duplex / Full-Duplex)
4. Introduction to Networks
5. Network Components
6. Network Criteria
7. Introduction to Layered Architecture
8. Protocols and Standards
9. Standards (De Facto / De Jure)
10. Standardization Organizations
11. Reference Models
12. OSI Reference Model
13. Physical Layer
14. Data Link Layer
15. Network Layer
16. Transport Layer
17. Session Layer
18. Presentation Layer
19. Application Layer
20. OSI Layer Summary
21. TCP/IP Reference Model
22. TCP/IP Application Layer
23. TCP/IP Transport Layer
24. TCP/IP Internet Layer
25. TCP/IP Network Access / Link Layer
26. OSI vs TCP/IP Reference Models
27. Transmission Media
28. Guided Transmission Media
29. Twisted Pair Cable
30. Coaxial Cable
31. Fiber Optic Cable
32. Unguided Transmission Media
33. Radio Waves
34. Microwaves
35. Infrared
36. Satellite Communication
37. Network Topology
38. Bus Topology
39. Star Topology
40. Ring Topology
41. Mesh Topology
42. Tree Topology
43. Hybrid Topology
44. Multiplexing

45. Frequency Division Multiplexing (FDM)
46. Time Division Multiplexing (TDM)
47. Wavelength Division Multiplexing (WDM)
48. Circuit Switching
49. Packet Switching
50. Datagram Packet Switching
51. Virtual Circuit Packet Switching
52. Circuit vs Packet Switching
53. Goals of Networks
54. Applications of Networks
55. LAN — Local Area Network
56. Wired LAN
57. Wireless LAN
58. LAN Interconnection Devices
59. Repeater
60. Hub
61. Bridge
62. Switch
63. Router
64. Gateway
65. Wireless Access Point
66. Interconnection Devices — Final Comparison
67. Complete PDU Journey
68. Encapsulation
69. Decapsulation
70. Most Important Definitions
71. High-Priority Differences
72. Final Revision Flow

1. Data Communication Concepts

1.1 Data Communication

Data communication is the exchange of data or information between two or more devices through a transmission medium, using a set of communication rules called protocols.



Fig. 1 — Basic data communication system, governed throughout by a shared PROTOCOL

Five components of data communication:

- Sender
- Receiver
- Message
- Transmission Medium
- Protocol

> **Why this matters:** Every networking concept in this unit is really an elaboration of one of these five components — layers refine the *protocol*, media chapters refine the *transmission medium*, and topology/switching chapters describe how many senders and receivers share that medium. Keeping this five-part picture in mind makes the rest of the unit easier to slot into place.

1.2 Sender

The sender is the device that generates and sends the data. Examples: Computer, Smartphone, Server, Sensor.

> **Example:** When you send an email, your computer/mobile acts as the sender.

1.3 Receiver

The receiver is the device that receives the transmitted data. Examples: Computer, Smartphone, Server, Printer.

> **Example:** The recipient's computer receiving an email is the receiver.

1.4 Message

The message is the actual information being communicated. It may be: Text, Numbers, Images, Audio, Video, Files.

1.5 Transmission Medium

A transmission medium is the physical or wireless path through which data travels from sender to receiver.

1.5.1 Guided Media

- Twisted Pair
- Coaxial Cable
- Fiber Optic

1.5.2 Unguided Media

- Radio Waves
- Microwaves
- Infrared
- Satellite

> **Note:** Guided media use a physical/wired path (signal is "guided" along a cable); unguided media broadcast signals through free space (air, vacuum). This split is expanded fully in Sections 27–36.

1.6 Protocol

A protocol is a set of rules that controls communication between devices. It defines:

- How data is formatted
- How data is transmitted
- How data is received
- How errors are handled

- How devices identify each other

Examples: TCP, IP, HTTP, HTTPS, FTP, SMTP, DNS.

> **Analogy:** Think of a protocol like the rules of a phone call — you say "hello," identify yourself, take turns speaking, and say "goodbye." Devices need the same kind of shared etiquette to exchange data correctly.

2. Characteristics of Data Communication

A good data communication system should provide:

2.1 Delivery

Data must reach the correct destination.

2.2 Accuracy

Data must arrive without errors or corruption.

2.3 Timeliness

Data should arrive within the required time. Important for: Video conferencing, Voice calls, Online gaming.

2.4 Jitter

Jitter is the variation in packet arrival time.

High jitter causes:

- Distorted voice
- Video freezing
- Poor real-time communication

> **Note:** Timeliness is about the average delay being low enough; jitter is about that delay being *consistent*. A link can have low average delay but still perform badly for a call if the delay bounces around a lot — real-time apps use "jitter buffers" to smooth this out before playback.

3. Data Flow

Data flow describes the direction in which data travels between communicating devices. There are three types: Simplex, Half-Duplex, Full-Duplex.

3.1 Simplex

In simplex communication, data flows in only one direction. A can transmit, but B cannot transmit back.



Fig. 2 — Simplex data flow: one way only

Examples: Keyboard → Computer, Television broadcasting, Traditional one-way radio broadcasting.

Advantages: Simple, Low cost, Easy implementation.
Disadvantage: No return communication is possible.

3.2 Half-Duplex

In half-duplex communication, data can flow in both directions, but not simultaneously.



Fig. 3 — Half-duplex data flow: both directions, only one at a time

Example: Walkie-talkie — one person talks while the other listens.

Advantage: Two-way communication is possible.
Disadvantage: Both devices cannot communicate simultaneously.

3.3 Full-Duplex

In full-duplex communication, data can flow in both directions simultaneously.



Fig. 4 — Full-duplex data flow: both directions, at the same time

Examples: Telephone call, Video call.
Advantage: Maximum communication efficiency.

Extra note: Full-duplex is usually achieved either by using separate transmit/receive wires or frequencies (so both directions physically never collide), or, in modern Ethernet switches, by giving each device its own dedicated link so there is no shared medium to contend for at all.

Quick comparison — Simplex vs Half-Duplex vs Full-Duplex

Simplex	Half-Duplex	Full-Duplex
One direction	Both directions	Both directions
Not simultaneous	Not simultaneous	Simultaneous
Keyboard / TV	Walkie-talkie	Telephone

4. Introduction to Networks

4.1 Computer Network

A computer network is a collection of interconnected computers and devices that communicate and share resources.

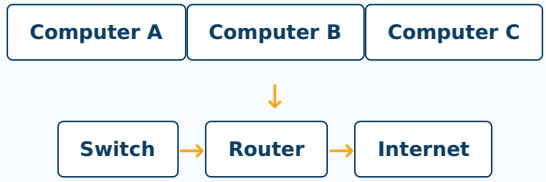


Fig. 5 — A basic computer network: end devices connect through a switch, which connects through a router to the wider Internet

4.2 Main Objectives of Networking

Networks are designed to provide:

- Resource sharing
- Communication
- Information sharing
- Reliability
- Scalability
- Cost reduction
- Distributed processing

> **Note:** These objectives reappear almost word-for-word in Section 53 ("Goals of Networks") — that section expands each one with concrete real-world examples, so treat Section 4.2 as the definition list and Section 53 as the applied version.

5. Network Components

A network generally contains:

5.1 End Devices

Devices that generate or receive data. Examples: Computer, Laptop, Smartphone, Printer, Server, IoT device.

5.2 Intermediary Devices

Devices that forward or manage network traffic. Examples: Switch, Router, Bridge, Repeater, Access Point.

5.3 Transmission Media

The path through which data travels. Examples: Ethernet cable, Fiber, Radio waves.

5.4 Network Protocols

Rules governing communication. Examples: TCP, IP, HTTP, DNS.

> **Extra note:** A handy way to remember the difference between "end devices" and "intermediary devices": end devices are the *source or destination* of data (they create or consume it), while intermediary devices only *move data along* without being the final destination. Sections 58–66 cover intermediary devices in full depth.

6. Network Criteria

A network is generally evaluated based on:

6.1 Performance

Performance indicates how efficiently a network transfers data.

Important parameters: Bandwidth, Throughput, Delay, Latency.

6.1.1 Bandwidth

Maximum data-carrying capacity of a communication channel. Measured in: bps, Kbps, Mbps, Gbps.

6.1.2 Throughput

Actual amount of data successfully transferred per unit time.

> **NOTE:** Bandwidth is the maximum theoretical capacity of a channel; throughput is the actual achieved rate, and it is always ≤ bandwidth in practice.

> **Example:** An ISP may advertise "100 Mbps bandwidth," but due to network congestion, signal loss, or server-side limits, the actual throughput you experience during a download might only be 60–80 Mbps.

6.1.3 Delay and Latency

Delay (or latency) is the time taken for a bit of data to travel from sender to receiver. It has several components worth knowing for exams:

- **Propagation delay** — time for a signal to physically travel across the medium (depends on distance and medium speed).
- **Transmission delay** — time to push all the bits of a packet onto the link (depends on packet size and bandwidth).
- **Queuing delay** — time a packet waits in a router/switch buffer before being forwarded.
- **Processing delay** — time a router takes to examine a packet's header and decide where to forward it.

6.2 Reliability

Reliability indicates the ability of a network to operate correctly and recover from failures. It depends on: Failure frequency, Recovery time, Error rate, Availability.

6.3 Security

Security protects Data, Devices, Network resources and Users from unauthorized access and attacks.

> **Extra note:** Security is commonly broken down into the "CIA triad": **Confidentiality** (only authorized parties can read the data), **Integrity** (data isn't altered in transit), and **Availability** (the network/service stays usable). This triad is a very common short-answer exam question.

7. Introduction to Layered Architecture

7.1 Layered Architecture

Layered architecture divides the complicated networking process into a number of smaller, independent layers.



Fig. 6 — The seven-layer model (top = Application, bottom = Physical)

Each layer performs a specific set of functions.

7.2 Need for Layered Architecture

7.2.1 Reduces complexity

Networking is divided into manageable parts.

7.2.2 Modularity

Each layer can perform a specific task.

7.2.3 Standardization

Different manufacturers can develop compatible systems.

7.2.4 Easy troubleshooting

Network problems can be isolated to a particular layer.

7.2.5 Easy modification

Changes to one layer can often be made without redesigning the entire architecture.

7.2.6 Interoperability

Different devices and systems can communicate using standardized interfaces and protocols.

> **Extra note — memory trick:** "Each layer only talks to the layer directly above and below it, using a well-defined interface." That single sentence explains modularity, easy troubleshooting, and easy modification all at once — a layer can be replaced or upgraded (e.g. swapping Wi-Fi for Ethernet) as long as it keeps offering the same service to the layer above it.

8. Protocols and Standards

8.1 Protocol

A protocol is a set of rules governing communication between network devices. A protocol defines: Syntax, Semantics, Timing.

8.2 Syntax

Syntax defines the format or structure of data.

> **Example:** It specifies where headers, addresses and data fields appear.

8.3 Semantics

Semantics defines the meaning of each field in a communication message.

> **Example:** In IP, semantics defines that a particular field represents the "destination address," and dictates what a router should do with that value — it's not enough to know a field's location; you must know what it *means*.

8.4 Timing

Timing defines:

- When data should be transmitted
- How quickly data should be transmitted
- How much time should be allowed between transmissions

> **Exam tip:** Syntax = "what," Semantics = "what it means," Timing = "when." A favorite exam question just asks you to classify a scenario into one of these three.

9. Standards

9.1 Standard

A standard is an agreed specification that ensures compatibility and interoperability between different devices and systems.

9.2 De Facto Standard

A de facto standard becomes a standard because it is widely used and accepted. It is not necessarily formally approved. Meaning: Standard by practice.

> **Example:** The QWERTY keyboard layout and the PDF file format both became de facto standards through widespread adoption before formal bodies ratified versions of them.

9.3 De Jure Standard

A de jure standard is formally approved by a recognized standardization organization. Meaning: Standard by law / formal approval.

> **Example:** The OSI model is a de jure standard because it was formally defined and published by ISO.

10. Standardization Organizations

10.1 ISO — International Organization for Standardization

Important contribution: OSI Reference Model.

10.2 IEEE — Institute of Electrical and Electronics Engineers

Important networking standards include: Ethernet, Wi-Fi.

10.3 IETF — Internet Engineering Task Force

Develops and maintains many Internet standards and protocols. Examples include standards related to: TCP, IP, HTTP, DNS.

10.4 ITU-T — International Telecommunication Union – Telecommunication Standardization Sector

Develops standards for telecommunications and related technologies.

> **Extra note:** IEEE standard numbers are frequently tested — e.g. IEEE 802.3 = Ethernet, IEEE 802.11 = Wi-Fi, IEEE 802.15 = Bluetooth/PAN. Knowing a couple of these numbers can help you answer "identify the standard" questions quickly.

11. Reference Models

A reference model provides a conceptual framework for describing network communication.

Two major reference models are: OSI Reference Model, TCP/IP Reference Model.

> **Note:** A reference model is not itself a protocol — it doesn't move any bits. It's a blueprint that tells protocol designers which job belongs at which layer, so that protocols built by different vendors and organizations still fit together correctly.

12. OSI Reference Model

12.1 Definition

OSI stands for Open Systems Interconnection. It was developed by ISO. The OSI model divides network communication into seven layers.



Fig. 7 — The seven OSI layers, 7 (Application) down to 1 (Physical)

> **Memory aid (mnemonic):** "**A**ll **P**eople **S**eem **T**o **N**eed **D**ata **P**rocessing" — Application, Presentation, Session, Transport, Network, Data Link, Physical (top to bottom). Reverse it — "**P**lease **D**o **N**ot **T**hrow **S**ausage **P**izza **A**way" — to go bottom to top.

13. Physical Layer

13.1 Definition

The Physical Layer is the lowest layer of the OSI model. It is responsible for transmitting raw bits over the physical communication medium.

13.2 Functions

13.2.1 Bit Transmission

Converts bits into signals and transmits them.

13.2.2 Data Rate

Defines the transmission rate.

13.2.3 Physical Topology

Defines how devices are physically connected.

13.2.4 Transmission Mode

Deals with simplex, half-duplex and full-duplex communication.

13.2.5 Physical Interfaces

Defines characteristics of connectors, cables, signal levels, etc.

13.3 Devices

Examples: Repeater, Hub, Cables, Connectors.

PDU: `Bits`

> **Note:** The Physical Layer doesn't understand addresses or meaning at all — it just deals with voltages, light pulses, or radio signals representing 0s and 1s. Any device that operates purely on raw signals (a repeater or hub) is, by definition, a Physical Layer device.

14. Data Link Layer

14.1 Definition

The Data Link Layer provides node-to-node delivery of data. It converts packets received from the Network Layer into frames.

14.2 Functions

14.2.1 Framing

Divides data into manageable units called frames.

14.2.2 Physical Addressing

Uses MAC addresses.

14.2.3 Error Detection

Detects errors in transmitted frames.

14.2.4 Flow Control

Controls the rate of data transmission between neighboring devices.

14.2.5 Access Control

Determines which device can access a shared medium.

14.3 Devices

Examples: Switch, Bridge. PDU: `Frame`

> **Extra note:** The Data Link Layer is often split conceptually into two sub-layers: **LLC (Logical Link Control)**, which handles flow control and error checking, and **MAC (Media Access Control)**, which handles addressing and controlling access to the shared medium. This is why physical hardware addresses are called "MAC addresses."

15. Network Layer

15.1 Definition

The Network Layer provides source-to-destination delivery across multiple interconnected networks.

15.2 Functions

15.2.1 Logical Addressing

Uses logical addresses such as IP addresses.

15.2.2 Routing

Determines the best path for packets.

15.2.3 Forwarding

Forwards packets toward their destination.

15.2.4 Internetworking

Connects different networks.

15.3 Device

Device: Router. PDU: Packet

> **Note — Routing vs Forwarding:** These two terms are often confused. **Routing** is the control-plane process of building and maintaining the routing table (deciding *what* the best paths are, often using routing protocols). **Forwarding** is the data-plane process of actually moving each incoming packet out the correct interface based on that table. Routing happens occasionally in the background; forwarding happens for every single packet.

16. Transport Layer

16.1 Definition

The Transport Layer provides process-to-process communication. It ensures that data is delivered to the correct application/process.

16.2 Major Functions

16.2.1 Segmentation

Breaks large data into smaller units.

16.2.2 Reassembly

Reconstructs the data at the destination.

16.2.3 Flow Control

Controls sender transmission rate.

16.2.4 Error Control

Provides mechanisms for reliable delivery.

16.2.5 Connection Control

Manages connection-oriented or connectionless communication.

16.3 Important Protocols

16.3.1 TCP — Transmission Control Protocol

Provides: Reliable delivery, Connection-oriented communication, Flow control, Congestion control, Error recovery.

16.3.2 UDP — User Datagram Protocol

Provides: Connectionless communication, Low overhead, No delivery guarantee.

PDU — TCP: Segment PDU — UDP: Datagram

> **Extra note:** "Process-to-process" delivery is what distinguishes the Transport Layer from the Network Layer's "host-to-host" delivery. An IP address gets a packet to the right *computer*; a TCP/UDP **port number** then gets the data to the right *application* running on that computer (e.g. port 80/443 for a web browser, port 25 for mail).

> **Example:** TCP is used for web browsing and file transfer, where every byte must arrive correctly and in order. UDP is used for live video calls and online gaming, where a dropped packet is preferable to the delay caused by waiting for retransmission.

17. Session Layer

17.1 Definition

The Session Layer establishes, manages and terminates communication sessions between applications.

17.2 Functions

17.2.1 Session Establishment

Starts a communication session.

17.2.2 Session Management

Maintains the session.

17.2.3 Session Termination

Ends the session.

17.2.4 Synchronization

Provides synchronization points during communication.

> **Note:** Synchronization points act like checkpoints in a long transfer — if the connection drops, communication can resume from the last checkpoint instead of starting over from scratch.

18. Presentation Layer

18.1 Definition

The Presentation Layer is responsible for the translation and representation of data.

18.2 Functions

18.2.1 Translation

Converts data between different formats.

18.2.2 Encryption

Provides encryption/decryption functions conceptually.

18.2.3 Compression

Reduces the size of data.

> **Example:** Converting between character encoding schemes (e.g. ASCII to Unicode), encoding an image as JPEG, or compressing a file as ZIP before it is sent are all Presentation Layer style functions — the data's *meaning* stays the same, but its *representation* changes.

19. Application Layer

19.1 Definition

The Application Layer provides network services directly to user applications.

19.2 Functions

Provides services such as: Web access, Email, File transfer, Remote access, Name resolution.

Examples: HTTP, HTTPS, FTP, SMTP, DNS, SSH.

> **Note:** This is the only layer that directly interfaces with the end user's software (browsers, mail clients, etc.). It is *not* the application itself, but the network-facing services and protocols that the application relies on.

20. OSI Layer Summary

OSI layer summary

Layer	Name	Main Function	PDU
7	Application	Network services	Data
6	Presentation	Translation, encryption, compression	Data
5	Session	Session management	Data
4	Transport	Process-to-process delivery	Segment
3	Network	Routing	Packet
2	Data Link	Framing / MAC	Frame
1	Physical	Bit transmission	Bits

Extra note: Layers 5–7 (Session, Presentation, Application) are often grouped together as the "upper layers" or "host layers" because they deal with the meaning and formatting of data for the user. Layers 1–4 are the "data flow layers" concerned with actually moving bits from one machine to another. This grouping is exactly why the TCP/IP model (Section 21) collapses 5–7 into one single Application layer.

21. TCP/IP Reference Model

The TCP/IP model is the architecture associated with the Internet protocol suite.

The commonly presented four-layer model is:

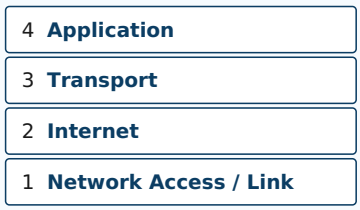


Fig. 8 — The four TCP/IP layers, Application down to Network Access / Link

Note: Unlike OSI, which was designed first as a theoretical model, TCP/IP was designed around the protocols that were already being built for the early Internet (ARPANET). This is why TCP/IP is described as a practical, "protocol-first" architecture rather than a pure reference model.

22. TCP/IP Application Layer

The Application Layer combines functions corresponding broadly to the OSI: Application, Presentation, Session.

Examples: HTTP, HTTPS, FTP, SMTP, DNS, SSH.

23. TCP/IP Transport Layer

Responsible for process-to-process communication.

Protocols: TCP, UDP.

Functions include: Segmentation, Flow control, Error control, Connection management.

24. TCP/IP Internet Layer

Responsible for communication between different networks.

Main protocol: IP.

Functions: Logical addressing, Routing, Packet forwarding.

Other protocol example: ICMP.

Example: ICMP is the protocol behind the "ping" command — it's used to test reachability and report errors (like "destination unreachable") rather than to carry user data.

25. TCP/IP Network Access / Link Layer

Responsible for communication over the local network and physical medium. It combines functions corresponding broadly to the OSI Data Link Layer and OSI Physical Layer.

Examples: Ethernet, Wi-Fi.

26. OSI vs TCP/IP Reference Models

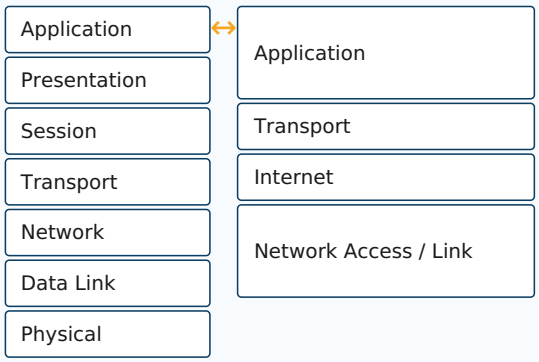


Fig. 9 — Mapping OSI's seven layers onto the TCP/IP four-layer model

OSI vs TCP/IP — feature comparison

Feature	OSI	TCP/IP
Number of layers	7	4 (commonly presented)
Developed by	ISO	Internet research / engineering community
Nature	Reference model	Protocol suite / architecture
Session Layer	Separate	Included in Application
Presentation Layer	Separate	Included in Application
Network Layer	Network	Internet
Data Link + Physical	Separate layers	Combined into Network Access / Link
Practical Internet usage	Primarily conceptual	Foundation of the Internet

> **NOTE:** Most important difference: OSI is mainly a reference model, whereas TCP/IP represents the actual architecture and protocols used by the Internet.

27. Transmission Media

27.1 Definition

Transmission media is the path through which signals travel from sender to receiver.

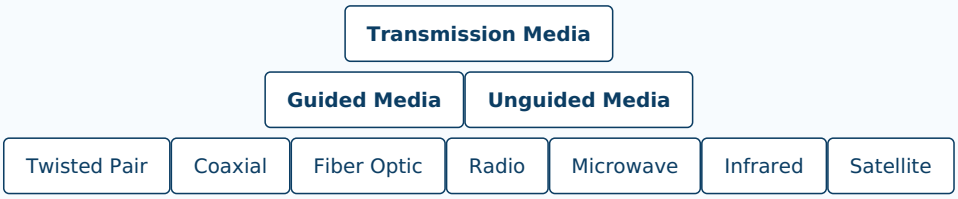


Fig. 10 — Two major categories of transmission media

28. Guided Transmission Media

In guided media, signals travel through a physical medium.

Types: Twisted Pair Cable, Coaxial Cable, Fiber Optic Cable.

29. Twisted Pair Cable

29.1 Definition

Twisted pair consists of two insulated copper wires twisted together to reduce electromagnetic interference.

29.2 Types

29.2.1 UTP — Unshielded Twisted Pair

No additional metallic shielding.

29.2.2 STP — Shielded Twisted Pair

Uses additional shielding to reduce interference.

29.3 Advantages

- Low cost
- Easy installation
- Flexible
- Widely available

29.4 Disadvantages

- Limited distance
- More susceptible to interference than fiber
- Lower capacity than fiber in many applications

> **Extra note:** The wires are twisted specifically to cancel out electromagnetic interference — each wire picks up roughly the same noise, and because the two wires carry inverted signals, the noise cancels out when the receiver takes the difference between them. This is the same principle used in balanced audio cables. Common categories you'll see in the field include Cat5e, Cat6, and Cat6a, each supporting higher speeds and longer reliable distances than the last.

30. Coaxial Cable

30.1 Definition

Coaxial cable consists of a central conductor surrounded by insulation and shielding.



Fig. 11 — Cross-section of a coaxial cable, from center outward

30.2 Advantages

- Better shielding
- Good bandwidth
- Less interference than basic twisted pair

30.3 Applications

- Cable television
- Broadband
- Older Ethernet systems

31. Fiber Optic Cable

31.1 Definition

Fiber optic cable transmits data using light signals through optical fiber.

31.2 Structure

Main components: Core, Cladding, Protective coating.

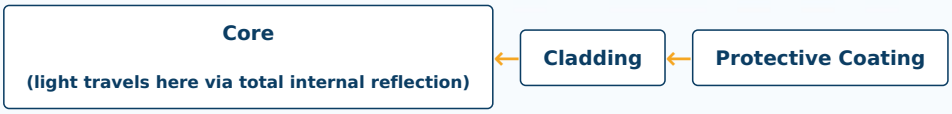


Fig. 12 — Structure of a fiber optic cable, core outward to protective coating

31.3 Working

Light travels through the core due to the principle of total internal reflection. Data is converted into light pulses and transmitted through the fiber.

31.4 Types

31.4.1 Single-Mode Fiber

Small core, one primary propagation mode, suitable for long distances, high bandwidth.

31.4.2 Multimode Fiber

Larger core, multiple propagation modes, generally used for shorter distances.

31.5 Advantages

- Very high bandwidth
- Long-distance transmission
- Low attenuation
- Immune to electromagnetic interference
- Lightweight

31.6 Disadvantages

- Higher installation cost
- Difficult to splice
- Requires specialized equipment

> **Extra note:** "Total internal reflection" happens because the cladding has a lower refractive index than the core — light hitting the core-cladding boundary at a shallow enough angle reflects entirely back into the core instead of escaping, letting it travel long distances with very little signal loss. This is also why fiber is immune to electromagnetic interference: it carries light, not electrical current, so it is unaffected by nearby power lines or radio signals.

32. Unguided Transmission Media

In unguided media, signals travel through free space.

Types: Radio waves, Microwaves, Infrared, Satellite communication.

33. Radio Waves

Radio waves are electromagnetic waves used for wireless communication.

Applications: Wi-Fi, Bluetooth, Radio broadcasting, Mobile communication.

Advantages: Wireless, Can cover large areas, Supports mobility.

Disadvantages: Interference, Security concerns, Signal attenuation.

> **Note:** Radio waves are typically omnidirectional — they spread out in all directions from the antenna, which is exactly what allows mobile devices to connect without needing to be precisely aimed. This omnidirectional nature is also why they are more prone to interference and eavesdropping than directional links.

34. Microwaves

Microwaves are electromagnetic waves generally used for directional communication.

Applications: Cellular communication, Satellite communication, Point-to-point wireless links.

Characteristics: High frequency, Directional, Often requires line-of-sight.

> **Example:** The dish antennas seen on rooftops or towers for point-to-point microwave links must be precisely aimed at each other, since microwave signals travel in a fairly narrow, straight beam and are blocked by obstacles such as hills or buildings.

35. Infrared

Infrared communication uses infrared electromagnetic radiation.

Characteristics: Short range, Generally cannot pass through walls, Suitable for room-level communication.

Example: Remote control systems.

> **Note:** Because infrared cannot pass through solid walls, it is naturally confined to a single room — this is actually a security advantage in some settings, since a signal can't easily be intercepted from outside the room.

36. Satellite Communication

Satellite communication uses satellites to relay signals between distant locations.



Fig. 13 — Satellite relaying signals between two earth stations

Advantages: Very large coverage, Useful for remote areas, Suitable for broadcasting.

Disadvantages: High propagation delay, Expensive infrastructure, Atmospheric effects can affect communication.

> **Extra note:** The high propagation delay comes from sheer distance — a geostationary satellite orbits roughly 36,000 km above Earth, so a signal traveling up and back down again (even at the speed of light) takes a noticeable fraction of a second, which is why satellite phone calls often have a perceptible lag. Low-Earth-orbit (LEO) satellite systems fly much closer to reduce this delay.

37. Network Topology

37.1 Definition

Network topology describes the arrangement of devices and communication links in a network.

Types: Bus, Star, Ring, Mesh, Tree, Hybrid.

> **Note:** Topology can be discussed at two levels — **physical topology** (how cables/devices are literally laid out) and **logical topology** (how data actually flows, which can differ from the physical layout). For example, modern Ethernet is usually wired physically as a star but can logically behave like a bus.

38. Bus Topology

All devices share a common backbone.



— all connected along a single Shared Backbone Cable —

Fig. 14 — Bus topology

Advantages: Simple, Less cable, Low initial cost.

Disadvantages: Backbone failure affects communication, Difficult troubleshooting, Performance decreases with traffic, Collisions may occur.

39. Star Topology

Every device connects to a central device.

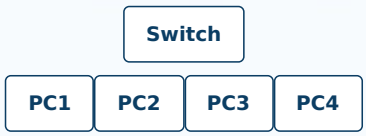


Fig. 15 — Star topology: each PC connects individually to the central switch

Advantages: Easy installation, Easy troubleshooting, Easy expansion, Failure of one cable generally affects only one device.

Disadvantage: Central device failure can affect the network.

Modern usage: Very common in Ethernet LANs.

40. Ring Topology

Each device connects to two neighboring devices.



Fig. 16 — Ring topology: the last device connects back to the first, closing the loop

Advantages: Organized communication, Predictable access in some implementations.

Disadvantages: Failure can affect communication depending on implementation, Adding/removing devices can be difficult.

Extra note: Classic ring networks (e.g. Token Ring) used a "token" that circulates around the ring — a device may only transmit when it holds the token, which avoids collisions but means a single broken link can disrupt the whole ring unless a dual-ring or bypass mechanism is used.

41. Mesh Topology

Devices have multiple connections to other devices.

41.1 Full Mesh

Every device is connected to every other device.

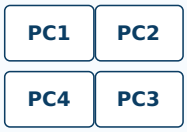


Fig. 17 — Full mesh topology (4 devices): every device links directly to every other device

CORRECTION: The number-of-links formula for a full mesh of n devices was garbled in the source. The correct formula is: **Number of links = n(n − 1) / 2**

Worked example: For n = 4 devices: links = 4(4-1)/2 = 4×3/2 = 6 links. For n = 6 devices: links = 6(5)/2 = 15 links. Notice how quickly this grows — this rapid growth is exactly why full mesh is expensive and rarely used at large scale.

Advantages: Highly reliable, Multiple paths, High fault tolerance.

Disadvantages: Expensive, Complex, Large number of connections.

42. Tree Topology

Tree topology provides a hierarchical arrangement of devices.

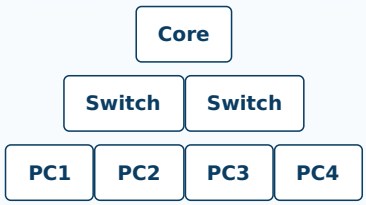


Fig. 18 — Tree topology: a hierarchical, branching layout rooted at the core

Advantages: Scalable, Hierarchical management, Suitable for large networks.

Disadvantages: Root/core failure can affect large portions, More complex than simple topologies.

> **Note:** A tree topology can be thought of as multiple star topologies connected together in a hierarchy — this is exactly how most real campus or enterprise networks are built, with access switches (stars) feeding up into distribution switches, which feed up into a core switch/router.

43. Hybrid Topology

Hybrid topology is a combination of two or more topologies.

Examples: Star + Bus, Star + Ring.

Advantage: Can combine the benefits of different topologies.

Disadvantage: More complex and expensive.

> **Extra note:** Almost every real-world large network (a university campus, a corporate office, the Internet itself) is a hybrid topology in practice — pure bus, ring, or full-mesh networks are mostly of academic/historical interest today, while star-of-stars and tree-like hybrids dominate real deployments.

Topology quick-reference table

Topology	Layout	Key strength	Key weakness
Bus	Shared backbone	Cheap, simple	Backbone failure = total outage
Star	Central hub/switch	Easy to manage/expand	Central device is a single point of failure
Ring	Closed loop	Organized, predictable access	Break in loop disrupts the ring
Mesh	Direct links between many/all nodes	Very fault-tolerant	Expensive, hard to scale
Tree	Hierarchical branches	Scalable, organized	Core failure affects many nodes
Hybrid	Combination	Flexible, best-of-both	Complex to design/manage

44. Multiplexing

44.1 Definition

Multiplexing is the technique of combining multiple signals and transmitting them over a single shared communication channel.

Multiplexer (MUX): Combines multiple input signals.

Demultiplexer (DEMUX): Separates the signals at the receiving end.

> **Note:** Multiplexing exists because physical links (a fiber, a cable, a slice of radio spectrum) are expensive relative to the number of users who want to communicate. Instead of laying a separate cable for every conversation, multiplexing lets many independent signals share one physical channel efficiently.

45. Frequency Division Multiplexing — FDM

45.1 Definition

FDM divides the available frequency spectrum into multiple frequency bands. Each signal gets a separate frequency band. All signals can be transmitted simultaneously.



Fig. 19 — Frequency Division Multiplexing: all 4 signals travel simultaneously, each in its own frequency band (bands shown left→right along the frequency axis)

45.2 Working

Suppose four users need to transmit data. Instead of giving them the same frequency, User A gets f1, User B gets f2, User C gets f3, User D gets f4. The signals are combined and transmitted over the same medium. At the receiver, they are separated using filtering / demultiplexing.

45.3 Applications

- Radio broadcasting
- Television broadcasting
- Traditional telephone systems
- Analog communication

Extra note: Between adjacent frequency bands, a small unused buffer called a "guard band" is usually left to prevent signals from bleeding into each other and causing interference — this is a common follow-up exam question ("why is FDM not 100% spectrum-efficient?").

46. Time Division Multiplexing — TDM

46.1 Definition

TDM divides the communication channel according to time. Each signal gets a specific time slot.



Fig. 20 — Time Division Multiplexing: each signal gets a fixed time slot, in turn (reading left to right along the time axis)

46.2 Types

46.2.1 Synchronous TDM

Each device gets a fixed time slot. Even if a device has nothing to send, its slot may remain unused.

46.2.2 Statistical TDM

Time slots are assigned dynamically based on demand. This improves efficiency when traffic is bursty.

Example: In Synchronous TDM, imagine 4 phone lines each getting a fixed 1 ms slot every cycle — even if one caller is silent, that slot goes unused and is "wasted." Statistical TDM instead only allocates a slot to lines that actually have data to send, similar to how a good scheduler avoids wasting CPU time on idle processes.

47. Wavelength Division Multiplexing — WDM

47.1 Definition

WDM is a multiplexing technique used primarily in fiber optic communication. Different signals use different wavelengths of light. Each wavelength carries an independent data stream.

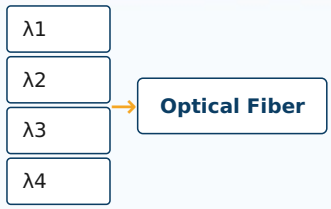


Fig. 21 — Wavelength Division Multiplexing: four independent data streams, each on its own wavelength, travel together down one fiber

Important relationship — FDM: Frequency | TDM: Time | WDM: Wavelength

Extra note: WDM is conceptually just FDM applied to light instead of radio/electrical signals — since wavelength and frequency of light are directly related (higher frequency = shorter wavelength), some textbooks describe WDM as "FDM for optical fiber." Dense WDM (DWDM) can carry dozens of separate wavelength channels on a single fiber, which is how modern long-haul Internet backbones achieve terabit-level capacity on one strand of glass.

48. Circuit Switching

48.1 Definition

Circuit switching establishes a dedicated communication path between sender and receiver before actual data transmission.

48.2 Phases of Circuit Switching



Fig. 22 — Phases of circuit switching

48.2.1 Phase 1: Connection Establishment

A dedicated path is established.

48.2.2 Phase 2: Data Transfer

Data travels through the established path.

48.2.3 Phase 3: Connection Release

The dedicated path is released after communication ends.

48.3 Characteristics

- Dedicated path
- Fixed resource allocation
- Connection-oriented
- Predictable transmission after setup

48.4 Advantages

- Dedicated bandwidth
- Predictable performance
- Low variation once established
- Suitable for continuous traffic

48.5 Disadvantages

- Setup delay
- Resources can remain unused
- Inefficient for bursty data
- A dedicated path may be unavailable during congestion

> **Example:** A traditional landline telephone call is the textbook example — dialing the number sets up a dedicated circuit through the telephone exchange, the entire call uses that fixed circuit exclusively even during silences, and hanging up releases it.

49. Packet Switching

49.1 Definition

Packet switching divides data into smaller units called packets. Each packet contains addressing / control information.

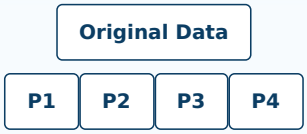


Fig. 23 — Packet switching: data divided into packets, each carrying its own addressing / control info and possibly routed independently

49.2 Working

- Data is divided into packets.
- Each packet receives required header information.
- Packets are transmitted through the network.
- Routers forward packets toward the destination.
- Data is reconstructed at the destination.

49.3 Advantages

- Efficient resource utilization
- Suitable for bursty data
- Multiple users can share links
- More scalable for computer networks

49.4 Disadvantages

- Variable delay
- Packets can be lost
- Packets may arrive out of order
- Congestion can occur

> **Note:** Because packets can be lost or arrive out of order, higher layers (particularly TCP at the Transport Layer) are responsible for detecting missing packets, requesting retransmission, and reassembling them in the correct order before handing data to the application.

50. Datagram Packet Switching

In a datagram network, each packet is treated independently. Packets can potentially take different paths (e.g. P1 and P3 via Path A, P2 via Path B).

The Internet's IP layer follows a datagram-style model.

> **Extra note:** Because each packet is routed independently and can take a different path, datagram networks are naturally resilient — if one path fails mid-transmission, later packets can simply be rerouted around the failure without needing to re-establish anything, unlike circuit switching.

51. Virtual Circuit Packet Switching

A logical path is established before packet transmission. Packets generally follow the established logical path. It combines some characteristics of circuit switching with packet switching.

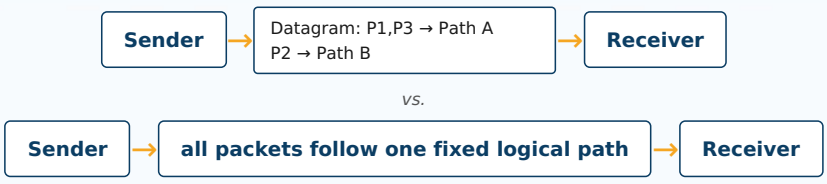


Fig. 24 — Datagram switching (top, independent paths) vs. Virtual Circuit switching (bottom, fixed logical path)

> **Example:** ATM (Asynchronous Transfer Mode) and MPLS are real-world technologies that use virtual circuits — a logical path is set up once, and all subsequent cells/packets for that connection follow it, giving more predictable performance than pure datagram routing while still sharing the underlying physical infrastructure.

52. Circuit vs Packet Switching

Circuit Switching vs Packet Switching

Feature	Circuit Switching	Packet Switching
Path	Dedicated	Shared
Setup	Required	Not necessarily required
Resource allocation	Dedicated	Dynamic
Efficiency	Lower for bursty traffic	Higher
Delay	More predictable after setup	Variable
Data division	Continuous stream	Packets
Example	Traditional telephone network	Internet

53. Goals of Networks

53.1 Resource Sharing

Networks allow users to share: Printers, Files, Storage, Applications, Internet connections, Servers.

53.2 Communication

Networks enable: Email, Messaging, Voice calls, Video calls, Collaboration.

53.3 Reliability

Networks can provide redundancy. If one component fails, another component may provide service.

53.4 Scalability

A network should be capable of supporting increasing numbers of: Users, Devices, Applications, Traffic.

53.5 Cost Reduction

Shared resources reduce infrastructure costs.

53.6 Distributed Processing

A large task can be divided among multiple computers.

Examples: Cloud computing, Distributed databases, Cluster computing.

54. Applications of Networks

54.1 Business

Email, File sharing, Cloud applications, Video conferencing, Enterprise systems.

54.2 Education

Online classes, E-learning, Digital libraries, Online examinations.

54.3 Banking

ATM, Online banking, Digital payments, Interbank communication.

54.4 Healthcare

Telemedicine, Medical databases, Remote monitoring, Electronic records.

54.5 Entertainment

Video streaming, Music streaming, Online gaming, Social media.

54.6 Government

E-governance, Online services, Digital records.

54.7 Industry

Automation, IoT, Remote monitoring, Industrial control.

55. LAN — Local Area Network

55.1 Definition

A LAN is a computer network covering a small geographical area, such as a: Room, Home, Office, Building, School, College campus.

55.2 Characteristics of LAN

- Limited geographical coverage
- High data rate
- Low latency
- Usually privately managed
- Resource sharing
- Relatively low error rate

> **Extra note — beyond LAN:** For exam completeness, it helps to know how LAN compares to its bigger siblings: a **MAN (Metropolitan Area Network)** covers a city, and a **WAN (Wide Area Network)** covers a country or the entire globe (the Internet is the largest WAN). As you move from LAN → MAN → WAN, coverage area increases but achievable data rate and latency performance generally get worse.

56. Wired LAN

56.1 Definition

A wired LAN connects devices using physical transmission media. The most common technology is Ethernet.

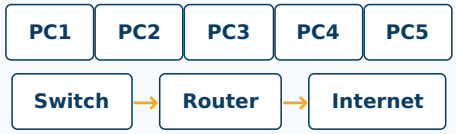


Fig. 25 — Typical wired LAN structure

56.2 Advantages

- High speed
- Reliable
- Low latency
- Stable performance
- Less affected by radio interference
- Better physical control of access to the medium

56.3 Disadvantages

- Requires cables
- Installation cost
- Limited mobility
- Cable management required

57. Wireless LAN

57.1 Definition

A Wireless LAN allows devices to communicate through wireless signals rather than physical network cables. The most common technology is Wi-Fi.

57.2 Basic WLAN Architecture

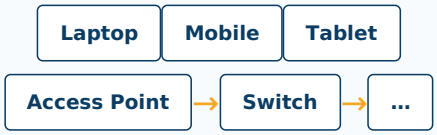


Fig. 26 — Basic wireless LAN architecture

57.3 Advantages

- Mobility
- Easy installation
- Reduced cabling
- Easy expansion
- Convenient access

57.4 Disadvantages

- Interference
- Security risks if poorly configured
- Variable performance
- Limited range
- Signal attenuation

> **Note:** Because wireless signals broadcast through open air, wireless LANs need extra security measures (e.g. WPA2/WPA3 encryption) that wired LANs can partly rely on physical access control for — anyone within range can potentially receive the radio signal, even if they can't decode it without the right credentials.

58. LAN Interconnection Devices

Important LAN interconnection devices:

- Repeater
- Hub
- Bridge
- Switch
- Router
- Gateway
- Wireless Access Point

> **Note:** A useful way to preview Sections 59–65: as you go from Repeater → Hub → Bridge → Switch → Router → Gateway, each device operates at an equal or higher OSI layer than the last, and each one adds more "intelligence" about the data it forwards — from raw signals, to MAC addresses, to IP addresses, to full protocol conversion.

59. Repeater

Definition: A repeater is a Physical Layer device used to regenerate signals so that communication can extend over a longer distance.

Main function: Signal regeneration. Layer: Physical Layer.

> **Example:** As a signal travels down a long Ethernet cable, it weakens (attenuates). A repeater receives the weakened signal, cleans it up, and retransmits it at full strength, effectively extending the maximum cable run.

60. Hub

Definition: A hub is a multiport repeater that sends incoming signals to multiple ports.

Characteristics:

- Physical Layer device
- No MAC-based intelligent forwarding
- Shared communication medium
- Incoming signals are repeated

Disadvantage: It creates a shared collision domain and is inefficient compared with modern switches.

> **Note — collision domain:** A collision domain is a section of a network where data packets can "collide" if two devices transmit at the same time. Because a hub blindly repeats every incoming signal out to every other port, all devices connected to a hub share one collision domain — this is precisely the inefficiency that switches (Section 62) were invented to fix.

61. Bridge

Definition: A bridge connects two or more LAN segments and uses MAC addresses to make forwarding decisions.

Layer: Data Link Layer.

Functions:

- Frame forwarding
- MAC address learning
- Traffic filtering
- Segmenting LAN traffic

> **NOTE:** A switch can be viewed conceptually as a multiport bridge.

62. Switch

Definition: A switch is a multiport networking device that forwards Ethernet frames using MAC addresses.

Layer: Primarily Data Link Layer.

62.1 MAC Address Table

A switch maintains a table like:

Example MAC address table

MAC Address	Port
MAC-A	1
MAC-B	2
MAC-C	3

It learns MAC addresses from the source addresses of incoming frames.

62.2 Switch Working

Suppose PC-A sends a frame to PC-B through the switch:



Fig. 27 — Switch forwarding: learns MAC-A on its port, checks table for MAC-B, forwards frame

The switch:

- Receives the frame.
- Reads source MAC address.
- Learns the source MAC and incoming port.
- Reads destination MAC address.
- Checks its MAC address table.
- Forwards the frame through the appropriate port if known.

62.3 Advantages

- Efficient forwarding
- Reduces unnecessary traffic
- Supports full-duplex communication
- Better performance than hubs
- Easy LAN expansion

> **Extra note:** What happens if the destination MAC address is *not* yet in the table? The switch "floods" the frame out of every port except the one it arrived on (much like a hub would, but only as a fallback) — once PC-B eventually replies, the switch learns MAC-B's port from that reply and adds it to the table, so future frames to PC-B are sent only out the correct port.

63. Router

Definition: A router is a network-layer device that connects different networks and forwards packets based on network-layer addressing.



Fig. 28 — Router connecting two LANs

Main Functions:

- Routing
- Packet forwarding
- Connecting different networks
- Maintaining routing information
- Separating broadcast domains in typical IP networks

Layer: Network Layer.

> **Note — broadcast domain:** A broadcast domain is the set of devices that receive a broadcast frame/packet sent by any one device. Switches do *not* separate broadcast domains (a broadcast still reaches every device on the switched LAN), but routers do — this is a common exam trap, since it's tempting to assume switches and routers behave the same way for both collision and broadcast domains.

64. Gateway

Definition: A gateway connects networks or systems that may use different protocols, architectures, or communication environments.

It may perform: Protocol conversion, Data translation, Network interconnection.

The exact function of a gateway depends on the context.

> **Example:** A "default gateway" on a home network is usually just the router, since it's the device that translates traffic between the local network's addressing and the wider Internet. In other contexts, "gateway" can mean a device that converts entirely different protocol stacks — e.g. bridging a modern IP network with a legacy proprietary industrial control system.

65. Wireless Access Point

Definition: A Wireless Access Point provides wireless devices with access to a network.

Functions:

- Provides Wi-Fi connectivity
- Connects wireless clients to LAN
- Manages wireless access
- Bridges wireless traffic to the wired network

66. Interconnection Devices — Final Comparison

LAN interconnection devices — final comparison

Device	Main Layer	Address Used	Main Function
Repeater	Physical	None	Regenerates signals
Hub	Physical	None	Repeats signals
Bridge	Data Link	MAC	Connects LAN segments
Switch	Data Link	MAC	Forwards frames
Router	Network	IP	Routes packets
Gateway	Depends	Depends	Protocol / system interconnection
Access Point	Mainly L2/L1	MAC / Wireless	Wireless connectivity

67. Complete PDU Journey

> **NOTE:** This is extremely important for exams and interviews.

Suppose you send data from one computer to another. At the sender, data moves down the layers; at the receiver, the process is reversed.

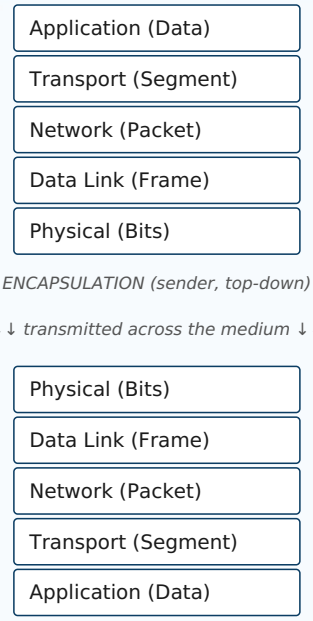


Fig. 29 — PDU journey: encapsulation at the sender, decapsulation at the receiver

68. Encapsulation

When data moves down the layers at the sender, each layer adds its own control information:

Layer	Adds
Application	Data
Transport	Segment
Network	Packet
Data Link	Frame
Physical	Bits

This is called Encapsulation.

> **Extra note:** More precisely, at each layer the data from the layer above is wrapped with a new header (and, at the Data Link layer, often a trailer too) — so a "frame" isn't a replacement for the packet, it's the packet *plus* a Data Link header/trailer wrapped around it. Each layer only needs to understand its own header; it treats everything from the layer above as opaque payload.

69. Decapsulation

At the receiver, headers are progressively removed as data moves upward through the layers: Bits → Frame → Packet → Segment → Data.

This is called Decapsulation.

> **Example:** Think of encapsulation like nested envelopes for a letter — the Application layer writes the "letter" (data), the Transport layer puts it in an envelope with a "room number" (port), the Network layer puts that inside a bigger envelope with a "building address" (IP address), and the Data Link layer puts that inside the outermost envelope with the "local courier's address" (MAC address). Decapsulation is simply opening each envelope in reverse order upon arrival.

70. Most Important Definitions to Memorize

70.1 Definition List

1. **Data Communication** — Exchange of data between two or more devices through a transmission medium using communication protocols.
2. **Computer Network** — A collection of interconnected devices that communicate and share resources.
3. **Protocol** — A set of rules governing communication between network devices.
4. **OSI Model** — A seven-layer reference model developed by ISO for standardizing and understanding network communication.
5. **TCP/IP Model** — A layered network architecture associated with the Internet protocol suite.
6. **Transmission Medium** — The path through which signals travel from sender to receiver.
7. **Multiplexing** — The technique of combining multiple signals for transmission over a shared communication channel.
8. **FDM** — Multiplexing technique in which different signals use different frequency bands.
9. **TDM** — Multiplexing technique in which different signals use different time slots.
10. **WDM** — Optical multiplexing technique in which different signals use different wavelengths of light.
11. **Circuit Switching** — Communication technique in which a dedicated path is established between sender and receiver before data transfer.
12. **Packet Switching** — Communication technique in which data is divided into packets and transmitted over shared network resources.
13. **LAN** — A network that connects devices within a limited geographical area.
14. **Switch** — A device that forwards frames using MAC addresses.
15. **Router** — A device that forwards packets between networks using network-layer addressing.

> **Bonus definitions worth adding to this list:**

16. **Bandwidth** — The maximum theoretical data-carrying capacity of a channel.
17. **Throughput** — The actual data successfully transferred per unit time (always ≤ bandwidth).
18. **Latency** — The total time delay for data to travel from sender to receiver.
19. **Encapsulation** — The process of wrapping data with header (and sometimes trailer) information as it passes down the layers at the sender.
20. **Collision Domain** — A network segment where two simultaneous transmissions can collide.
21. **Broadcast Domain** — The set of devices that receive a broadcast sent by any device within it; routers separate broadcast domains, switches do not.

71. High-Priority Differences

71.1 Simplex vs Half-Duplex vs Full-Duplex

Simplex	Half-Duplex	Full-Duplex
One direction	Both directions	Both directions
Not simultaneous	Not simultaneous	Simultaneous
Keyboard / TV	Walkie-talkie	Telephone

71.2 FDM vs TDM vs WDM

FDM	TDM	WDM
Frequency	Time	Wavelength
Frequency bands	Time slots	Light wavelengths
Radio / analog systems	Digital / shared systems	Fiber optics

71.3 Hub vs Switch vs Router

Hub	Switch	Router
Physical layer	Data Link	Network
Repeats signals	Forwards frames	Forwards packets
No MAC intelligence	Uses MAC	Uses IP / routing
Shared collision domain	Separate collision domains per port (typical Ethernet)	Separates networks / broadcast domains

71.4 Circuit vs Packet Switching

Circuit	Packet
Dedicated path	Shared resources
Fixed allocation	Dynamic allocation
Setup required	No dedicated circuit required
Predictable	Variable delay
Traditional telephone	Internet

71.5 OSI vs TCP/IP (recap)

OSI	TCP/IP
7 layers	4 layers (commonly presented)
Reference model	Protocol suite / architecture
Session & Presentation separate	Merged into Application

72. Final Revision Flow

Remember the entire unit in this order:

1. DATA COMMUNICATION
2. Sender + Receiver + Message + Medium + Protocol
3. Data Flow (Simplex / Half / Full Duplex)
4. COMPUTER NETWORK
5. Layered Architecture
6. Protocols + Standards
7. OSI MODEL: 7 Application → 6 Presentation → 5 Session → 4 Transport → 3 Network → 2 Data Link → 1 Physical
8. TCP/IP MODEL: Application → Transport → Internet → Network Access
9. OSI vs TCP/IP
10. TRANSMISSION MEDIA: Guided (Twisted Pair, Coaxial, Fiber) / Unguided (Radio, Microwave, Infrared, Satellite)
11. NETWORK TOPOLOGY: Bus, Star, Ring, Mesh, Tree, Hybrid
12. MULTIPLEXING: FDM, TDM, WDM
13. SWITCHING: Circuit Switching, Packet Switching
14. GOALS & APPLICATIONS
15. LAN: Wired LAN, Wireless LAN
16. LAN INTERCONNECTION DEVICES: Repeater, Hub, Bridge, Switch, Router, Gateway, Access Point

> **NOTE:** Use this flow as a last-minute revision checklist — if you can explain every line above out loud without looking, you know this unit.